

Brijodison

Engineering Notebook Element E

Engineering Service Learning

Application of STEM Principles

There are many ways to go through the STEM principles and practices. The STEM practices that we believed we used were hypothesizing, explaining based on evidence, designing and carrying out experiments, and developing. The STEM principles that we used were critical thinking, collaboration, communication and creativity. We believe that the ones we used were a very important part of creating our design. Throughout creating our design we believe that we reached all of these principles and practices.

The main STEM practices that we went through were as listed in the beginning: hypothesizing, explaining based on evidence, designing and carrying out experiments, and developing. In the beginning of our process we had a lot of valuable hypothesizing while coming up with ideas. Before we even started writing down our ideas we each went through many possible solutions to the problem and shared with each other. We then collectively came to agreements about what we should take and how. We then explained all of our solutions using the previous evidence we gained from Mr. Kiser's interview. We went through the requirements needed to complete the prompt and decided on which idea would best suit what we wanted to accomplish. After we finished we started to design and carry out our experiments. After we would create something we would test how functional it was on the cart and see where we could better our invention. Once we were satisfied with where we were at in our designing process we could continue to develop our design.

While doing this whole process we kept in mind our STEM principles that were as listed above: critical thinking, collaboration, communication, and creativity. During our project we have used a lot of critical thinking while trying to design our project. While brainstorming and coming

up with ideas we thought about what would effectively work and how we could convey that through our design. Throughout our project we used a lot of collaboration skills while working. We worked together to help each other understand what we were thinking and how we were trying to accomplish our project. We made sure that while any of us were trying something new we made sure to keep everyone included. We also used a lot of communication skills while working. Throughout our project we met together and explained what we were going to try to do for the day and if anybody has any questions. During our days together we also made sure that before we started any part of the project we made sure it was ok with everybody else. We believe that we communicated well with each other which made our process go by smoother. We also showed a lot of creativity working together trying to find a design that was effective. We allowed each other to present a point and how it would work and gave each other the creative freedom to come up with new ideas and designs.

On our “designathon” day we had the opportunity to share our design process and the STEM principles and practices we took with Vanderbilt students and graduates. The first person that we were able to share with was Graduate Student Jared Lawson. When he came to our table we explained to him the steps we took and the design matrix that we used. He gave us plenty of valuable feedback and information about our process. He was even able to help us with our developing part of our project. By walking us through new steps to take and new ideas. The second person that we got to share our process with was Sales force Peter Hermann. He agreed with the steps that we took and also gave us important information on how we could better our product.

Throughout our entire process we effectively used our STEM principles and practices which has helped and benefited us through our design process. We heard valuable feedback from our “designathon” which helped us and contributed to our project.

